Market Survey

ENERGY STORAGE: What's Driving The Growth?

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While lack of reliable grid power was a driver for traditional energy storage systems, ambitious renewable energy goals, growth in electric vehicle sales, and need for reduction in diesel usage will boost the sales of energy storage systems throughout the country

Energy storage systems are essential for large renewable energy systems to provide a clean, resilient energy supply through local generation. The technology continues to prove its value to grid operators around the world, who must manage the variable generation of solar and wind energy.

Several studies predict considerable near-term potential for stationary energy storage as its costs are falling. In fact, its cost could reduce to $200 per kilowatt-hour (half the current price) in 2020 and $160 per kilowatt-hour or less in 2025.

Besides, identifying the most economical projects and highest-potential customers for storage has become a priority for a diverse set of companies including power suppliers, grid operators, battery manufacturers, energy-storage integrators, and businesses having established relationships with prospective customers such as solar developers and energy-service companies. According to McKinsey, the global opportunity for storage could reach 1000GW in the next 20 years as the technology matures.

Significance of energy storage business

Energy storage systems absorb and release power when required. Major forms of energy storage systems include lithium-ion, lead-acid and moulten-salt batteries, as well as flow cells.

There are four major benefits of energy storage. First, it can be used to smooth out the flow of fluctuating power supply. Second, storage can be integrated into electricity systems so that if the main source of power fails, it provides a backup service. Third, storage can increase utilisation of power-generation or transmission and distribution assets, for example, by absorbing power that exceeds current demand. Fourth, in some markets, the cost of generating power is significantly cheaper at one point in time than another; storage can help mitigate fluctuating costs.

Historically, manufacturers, grid operators, independent power producers and utility providers have invested in energy-storage devices to meet their own needs or service their territory. As storage costs fall, ownership will broaden and many new business models will emerge.

Indian scenario

India is one of the fastest growing economies in the world. However, its annual per capita energy consumption of around 1222kWh is far below the level in other developing countries like Brazil and China. Therefore India needs a clean energy revolution in order to provide energy access to its 240 million households that still lack this basic farility.

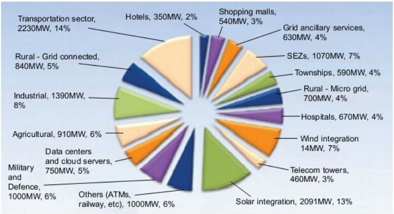

Fig. 1: India's energy storage market potential by 2020 (Source: India Energy Storage Alliance report)

Moreover, urban middle-class population of the country, exceeding 250 million, aspires for the same level of energy access and power quality as in developed countries through movements like Smart Cities and sustainable living.

Lack of reliable and quality power supply is a major obstacle to the development of manufacturing and industrial sector in India. Industries rely on diesel-powered generators, while households use inverters with batteries to deal with unscheduled power cuts, low voltages or variable frequency.

Given this scenario, energy storage systems can play a key role across all segments in India. India Energy Storage Alliance (IESA) estimates the Indian energy storage market to reach 16,445MW by 2020 (Fig. 1) and further up to 70GW by 2022. This huge market potential will attract huge investment.

India has the potential to not only